# KATHULAPALLI CHAKRADHAR REDDY

Enrollment No.: BT21CSE101 • B. Tech. Computer Science and Engineering
Visvesvaraya National Institute of Technology, Nagpur

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## EDUCATION

| Degree/Course | Institute /Board | CGPA/Percentage | Year |
|---|---|---|---|
| **B.Tech(CSE)** | Visvesvaraya National Institute of Technology, Nagpur | **7.02** | 2026 |
| CLASS XII (BIEAP) | Narayana Junior college, Vijayawada | **96.7%** | 2021 |
| CLASS X (SSC) | Dr. KKR's Gowtham International School, Vijayawada | **9.7 GPA** | 2019 |

## PROJECTS

**EdgeProbe: Automated CDN, DNS & Distributed Telemetry QA Framework** Github

- Architected an automated SDET verification framework validating edge caching compliance (RFC 9111), Anycast DNS routing latency, and analytical telemetry ingestion integrity with **34 hermetic regression tests passing in 0.5s**.
- Asserted HTTP cache contracts including **X-Cache HIT/MISS transitions**, TTL expiration, conditional **304 Not Modified revalidation via ETags**, and upstream **Fast Purge** cache invalidation.
- Engineered multi-resolver DNS Anycast latency probers against Google (8.8.8.8) and Cloudflare (1.1.1.1), enforcing strict **sub-150ms SLA quality gates**; automated SQLAlchemy warehouse checks detecting duplicates and schema drift.

— **Tools & technologies used:** Python, Pytest, FastAPI, SQLAlchemy, dnspython, HTTPX, Docker Compose, SQLite, PostgreSQL, GitHub Actions.

**Cache Poisoning Mitigation using Sliding Window and Multi-Agent Reinforcement Learning (SW-MARL)** Github

- Developed a security-aware cache management framework for Information-Centric Networking (ICN/NDN) using Multi-Agent Reinforcement Learning (MARL) to prevent adversarial cache pollution attacks.
- Designed a sliding-window temporal state representation enabling routers to learn adaptive cache admission and eviction policies; achieved **0.0% poisoned content**, **92.3% cache hit rate**, and automatic self-healing.
- Contributed as project lead under Prof. Nidhi Lal; research paper submitted to **ICICT 2026**.

— **Tools & technologies used:** Python, PyTorch, PettingZoo, RLlib, SimPy, NetworkX, Multi-Agent Reinforcement Learning, ICN/NDN.

**Enterprise Network Automation & Real-Time Monitoring Platform** Github

- Architected a multi-vendor automation and telemetry platform managing Cisco IOS routers & switches (Catalyst 3750, 7200) via Netmiko, FastAPI, and Ansible with zero-touch Jinja2 provisioning.
- Engineered automated root-cause diagnostic engine for interface errors and OSPF troubleshooting; containerized with Docker and verified via GitHub Actions CI/CD.

— **Tools & technologies used:** Python, FastAPI, Cisco IOS, Ansible, Terraform, AWS (VPC, EC2), Docker, Netmiko, SQLite, Pytest, GitHub Actions.

**Student Placement Management System** Github

- Developed a full-stack student placement platform with separate Student, Company, and Admin roles secured with JWT-based authentication and role-based access control (RBAC).
- Enforced centralized validation for duplicate applications, deadlines, and CGPA eligibility; designed RESTful APIs with Spring Boot, React, and MySQL via Spring Data JPA.

— **Tools & technologies used:** Java, Spring Boot, Spring Data JPA, React, MySQL, REST APIs, JWT, Maven, Git/GitHub.

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Java, JavaScript, TypeScript, HTML, CSS, SQL, Bash.

**SDET & Quality Engineering:** Pytest Framework, RFC 9111 HTTP Caching, dnspython (Anycast DNS), SQLAlchemy Testing, HTTPX Async Probing, Data Warehouse SQL Quality Gates, Performance SLA Benchmarking, Test Automation.

**Frameworks & Technologies:** Spring Boot, Spring Data JPA, FastAPI, React, Node.js, REST APIs, JWT, PyTorch, PettingZoo, RLlib, SimPy, NetworkX.

**Network & DevOps:** Cisco IOS, Ansible, Docker, Kubernetes, Terraform (IaC), Netmiko, GitHub Actions, CI/CD, Linux/Bash.

**Core Skills:** Data Structures & Algorithms, OOP, Deep Reinforcement Learning, Multi-Agent Systems, Network Security.

**Databases:** MySQL, PostgreSQL, MongoDB, SQLite.

**Tools & Cloud:** VS Code, Git, GitHub, Maven, AWS (EC2, VPC, S3), Postman, GNS3, EVE-NG.

## POSITIONS OF RESPONSIBILITY

- **Event Manager**, AXIS VNIT (Central India's Largest Annual Technical Festival) • Managed event operations, logistics, technical competitions, and team coordination.
- **Alumni Secretary**, Department of Computer Science and Engineering, VNIT Nagpur • Coordinated departmental alumni relations, student interaction sessions, and career workshops.